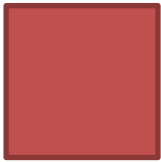


LESSON 7 : AREA AND VOLUME (A)

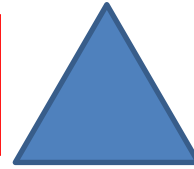
How to calculate the area of the following shapes?



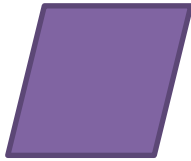
$$A = l^2$$



$$A = lw$$



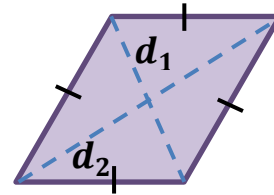
$$A = \frac{bh}{2}$$



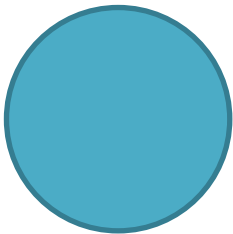
$$A = bh$$



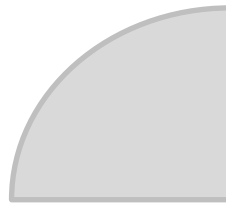
$$A = \frac{b_1 + b_2}{2} h$$



$$A = \frac{d_1 d_2}{2}$$



$$A = \pi r^2$$



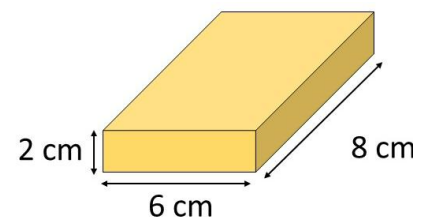
$$A = \pi r^2 \times \frac{\theta}{360}$$

Checkpoint 1

Volume of Prism and Cylinder

Volume of Prism = Base area $\times$ Height

Volume of Cylinder = $\pi r^2 h$

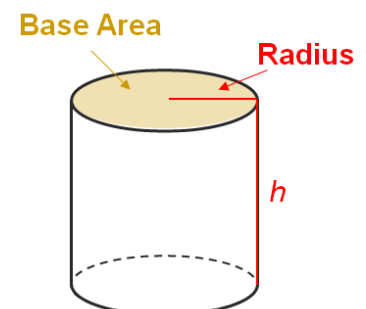


Total Surface Area of right Prism and Cylinder

TSA of right Prism

= Base area $\times$ 2 + Perimeter of Base $\times$ Height

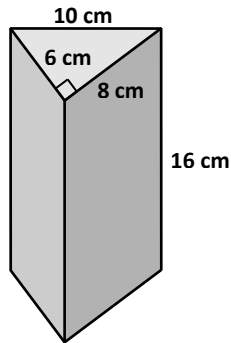
TSA of right Cylinder = $2\pi r^2 + 2\pi rh$



PRE S3 MATHS L7 – AREA AND VOLUME (A)

Find the volume and the total surface area of the following right solids (in terms of π if needed). (Q1-3)

1.



Base area

$$= \left(\frac{1}{2} \times 6 \times 8 \right) \text{ cm}^2$$

$$= 24 \text{ cm}^2$$

Volume of the prism

$$= 24 \times 16 \text{ cm}^3$$

$$= \underline{\underline{384 \text{ cm}^3}}$$

$$2 \times \text{base area}$$

$$= 2 \times 24 \text{ cm}^2$$

$$= 48 \text{ cm}^2$$

Total area of all lateral faces

$$= (6 + 8 + 10) \times 16 \text{ cm}^2$$

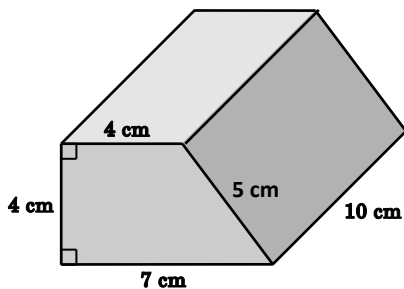
$$= 384 \text{ cm}^2$$

$\therefore$ Total surface area of the prism

$$= (48 + 384) \text{ cm}^2$$

$$= \underline{\underline{432 \text{ cm}^2}}$$

2.



Base area

$$= \left[\frac{1}{2} (4 + 7) \times 4 \right] \text{ cm}^2$$

$$= 22 \text{ cm}^2$$

Volume of the prism

$$= 22 \times 10 \text{ cm}^3$$

$$= \underline{\underline{220 \text{ cm}^3}}$$

$$2 \times \text{base area}$$

$$= 2 \times 22 \text{ cm}^2$$

$$= 44 \text{ cm}^2$$

Total area of all lateral faces

$$= (7 + 4 + 4 + 5) \times 10 \text{ cm}^2$$

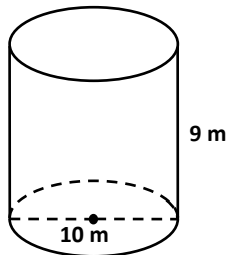
$$= 200 \text{ cm}^2$$

$\therefore$ Total surface area of the prism

$$= (44 + 200) \text{ cm}^2$$

$$= \underline{\underline{244 \text{ cm}^2}}$$

3.



Volume of the cylinder

$$= \left[\pi \left(\frac{10}{2} \right)^2 \times 9 \right] \text{ m}^3$$

$$= 225\pi \text{ m}^3$$

$$\text{Total surface area} = \left[90\pi + 2\pi \times \left(\frac{10}{2} \right)^2 \right] \text{ m}^2$$

$$= \underline{\underline{140\pi \text{ m}^2}}$$

4. The total surface area of a right cylinder and the circumference of its base are $1\,312\pi \text{ mm}^2$ and $32\pi \text{ mm}$ respectively. Find its

(a) height, (b) volume. (Give the answers in terms of π if necessary.)

(a) Let $r \text{ mm}$ be the base radius of the cylinder.

$\therefore$ Circumference of its base $= 32\pi \text{ mm}$

$$\therefore 2\pi r = 32\pi$$

$$r = \frac{32\pi}{2\pi}$$

$$= 16$$

$\therefore$ The base radius of the cylinder is 16 mm .

Let $h \text{ mm}$ be the height of the cylinder.

$\therefore$ Total surface area $= 1\,312\pi \text{ mm}^2$

$$\therefore 2\pi \times 16 \times h + 2\pi \times 16^2 = 1\,312\pi$$

$$32\pi h + 512\pi = 1\,312\pi$$

$$32\pi h = 800\pi$$

$$h = \frac{800\pi}{32\pi}$$

$$= 25$$

$\therefore$ The height of the cylinder is 25 mm .

(b) Volume of the cylinder $= \pi \times 16^2 \times 25 \text{ mm}^3$

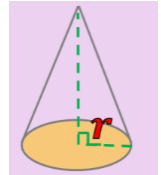
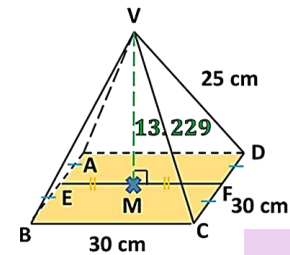
$$= \underline{\underline{6\,400\pi \text{ mm}^3}}$$

Checkpoint 2

Volume of Pyramid and Cone

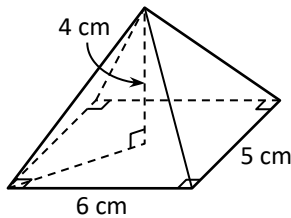
$$\text{Volume of Pyramid} = \frac{1}{3} \times \text{Base area} \times \text{Height}$$

$$\text{Volume of Cone} = \frac{1}{3} \pi r^2 h$$



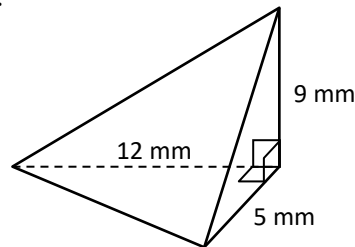
Find the volume of the following solids (in terms of π if needed). (Q5-7)

5.



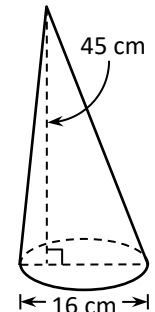
$$\begin{aligned} \text{Volume} &= \frac{1}{3} \times (6 \times 5) \times 4 \text{ cm}^3 \\ &= \underline{40 \text{ cm}^3} \end{aligned}$$

6.



$$\begin{aligned} \text{Volume} &= \frac{1}{3} \times \left(\frac{1}{2} \times 5 \times 12 \right) \times 9 \text{ mm}^3 \\ &= \underline{90 \text{ mm}^3} \end{aligned}$$

7.



$$\begin{aligned} \text{Volume} &= \frac{1}{3} \times \pi \times \left(\frac{16}{2} \right)^2 \times 45 \text{ cm}^3 \\ &= \underline{960\pi \text{ cm}^3} \end{aligned}$$

8. Lucy has eight identical cups in the shape of inverted right circular cones. The base diameter and the height of each cup are 9 cm and 14 cm respectively. Can the eight cups hold orange juice of volume 2 250 cm³? Explain your answer.

$$\begin{aligned} &\text{Total capacity of the eight cups} \\ &= 8 \times \frac{1}{3} \times \pi \times \left(\frac{9}{2} \right)^2 \times 14 \text{ cm}^3 \\ &= 2\,375 \text{ cm}^3, \text{ cor. to the nearest cm}^3 \\ &> 2\,250 \text{ cm}^3 \\ &\therefore \underline{\text{The eight cups can hold orange juice of volume}} \\ &\quad \underline{2\,250 \text{ cm}^3.} \end{aligned}$$

9. The figure shows a pyramid $VPQR$ whose height is VP . The base of the pyramid is a right-angled triangle PQR , where $PQ \perp QR$. It is given that $VQ = 7.5$ m, $PQ = 6$ m and $QR = 10$ m.

- (a) Find the height of the pyramid.
(b) Find the volume of the pyramid.

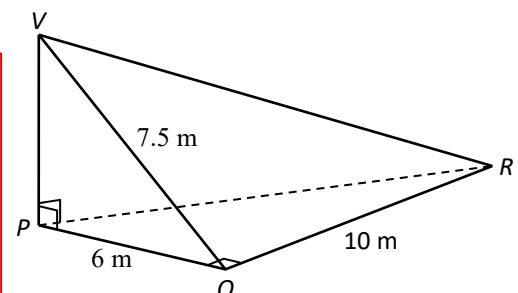
(a) In $\triangle VPQ$,

$$\begin{aligned} VP &= \sqrt{VQ^2 - PQ^2} \\ &= \sqrt{7.5^2 - 6^2} \text{ m} \\ &= 4.5 \text{ m} \end{aligned}$$

$\therefore$ The height of the pyramid is 4.5 m.

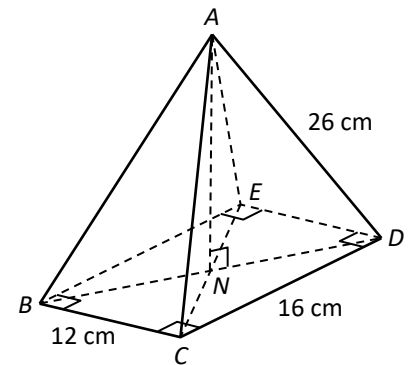
(b) Volume of the pyramid

$$\begin{aligned} &= \frac{1}{3} \times \text{area of } \triangle PQR \times VP \\ &= \frac{1}{3} \times \left(\frac{1}{2} \times 6 \times 10 \right) \times 4.5 \text{ m}^3 \\ &= \underline{45 \text{ m}^3} \end{aligned}$$



10. The figure shows a right pyramid $ABCDE$ with a rectangular base $BCDE$. The diagonals BD and CE intersect at N . $AD = 26$ cm, $BC = 12$ cm and $CD = 16$ cm.

- (a) Find the lengths of ND and AN .
(b) Find the volume of the pyramid.



(a) In $\triangle BCD$,

$$\begin{aligned} BD &= \sqrt{BC^2 + CD^2} \\ &= \sqrt{12^2 + 16^2} \text{ cm} \\ &= 20 \text{ cm} \end{aligned}$$

$$ND = \frac{1}{2} \times BD = \frac{1}{2} \times 20 \text{ cm} = \underline{10 \text{ cm}}$$

In $\triangle ADN$,

$$\begin{aligned} AN &= \sqrt{AD^2 - ND^2} \\ &= \sqrt{26^2 - 10^2} \text{ cm} \\ &= \underline{24 \text{ cm}} \end{aligned}$$

(b) Volume of the pyramid

$$\begin{aligned} &= \frac{1}{3} \times \text{area of rectangle } BCDE \times AN \\ &= \frac{1}{3} \times (16 \times 12) \times 24 \text{ cm}^3 \\ &= \underline{1536 \text{ cm}^3} \end{aligned}$$

11. The base of the pyramid in the figure is a right-angled triangle QRS , where $QR \perp RS$ and $RS = 6$ cm. The height PQ of the pyramid is 7 cm. If the volume of the pyramid is 28 cm^3 ,

- (a) find the area of $\triangle QRS$,
(b) find the lengths of QR and the slant edge PR .
(Give the answers correct to the nearest 0.1 cm if necessary.)

(a) Let $x \text{ cm}^2$ be the area of $\triangle QRS$.

$$\frac{1}{3} \times x \times 7 = 28$$

$$x = 12$$

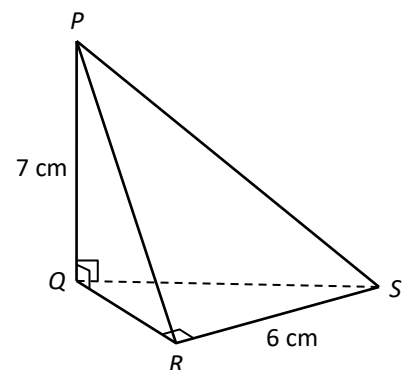
$\therefore$ The area of $\triangle QRS$ is 12 cm^2 .

(b) $\frac{1}{2} \times RS \times QR = 12 \text{ cm}^2$

$$\begin{aligned} QR &= \frac{2 \times 12}{6} \text{ cm} \\ &= \underline{4 \text{ cm}} \end{aligned}$$

In $\triangle PQR$,

$$\begin{aligned} PR &= \sqrt{PQ^2 + QR^2} \\ &= \sqrt{7^2 + 4^2} \text{ cm} \\ &= \underline{8.1 \text{ cm, cor. to the nearest 0.1 cm}} \end{aligned}$$



12. The figure shows a pyramid. The base of the pyramid is a regular hexagon with side length of 6 cm. The height of the pyramid is 12 cm. Find the volume of the pyramid.

$$3^2 + y^2 = 6^2$$

$$y^2 = 27$$

$$y = \sqrt{27}$$

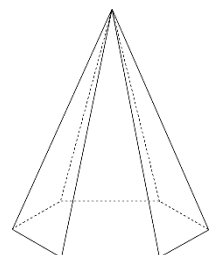
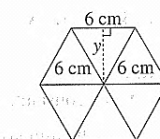
Base area

$$= \frac{1}{2} \times 6 \times \sqrt{27} \times 6$$

$$= 54\sqrt{3} \text{ cm}^2$$

$$\text{Volume} = \frac{1}{3} \times 54\sqrt{3} \times 12$$

$$= \underline{216\sqrt{3} \text{ cm}^3 (= 374 \text{ cm}^3)}$$



Checkpoint 3

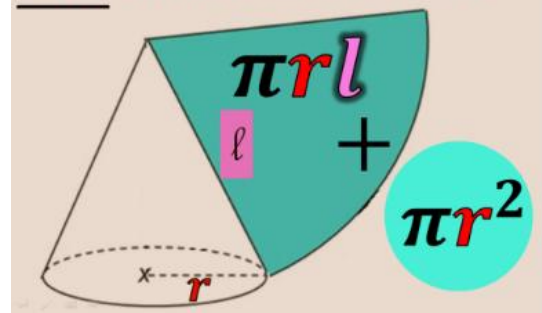
Curved surface area of Cone

$$= \pi r l$$

Total surface area of Cone

$$= \pi r^2 + \pi r l$$

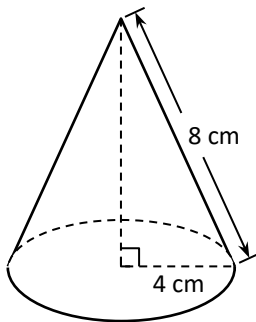
Total surface area of cone



For each right circular cone in Q13-15 find

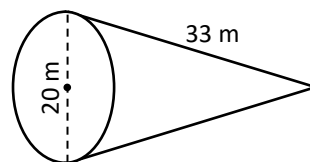
- (a) the curved surface area, (b) the total surface area. (Give the answers in terms of π .)

13.



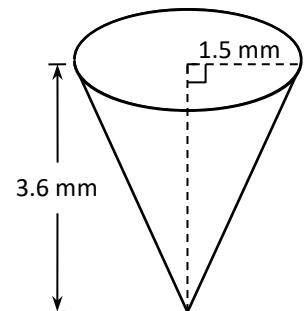
- (a) Curved surface area $= \pi \times 4 \times 8 \text{ cm}^2$
 $= \underline{32\pi \text{ cm}^2}$
 (b) Total surface area $= (32\pi + \pi \times 4^2) \text{ cm}^2$
 $= \underline{48\pi \text{ cm}^2}$

14.



- (a) Curved surface area $= \pi \times \frac{20}{2} \times 33 \text{ m}^2$
 $= \underline{330\pi \text{ m}^2}$
 (b) Total surface area $= \left[330\pi + \pi \times \left(\frac{20}{2}\right)^2 \right] \text{ m}^2$
 $= \underline{430\pi \text{ m}^2}$

15.



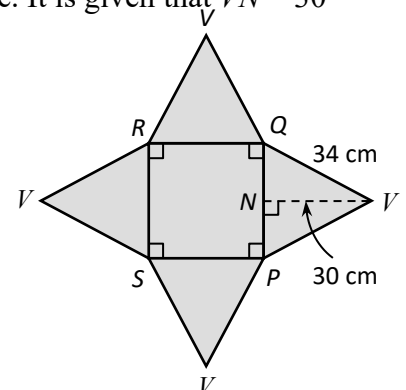
- (a) Let ℓ mm be the slant height of the cone.
 $1.5^2 + 3.6^2 = \ell^2$
 $\ell = \sqrt{1.5^2 + 3.6^2}$
 $= 3.9$
 $\therefore$ Curved surface area $= \pi \times 1.5 \times 3.9 \text{ mm}^2$
 $= \underline{5.85\pi \text{ mm}^2}$
 (b) Total surface area
 $= (5.85\pi + \pi \times 1.5^2) \text{ mm}^2$
 $= \underline{8.1\pi \text{ mm}^2}$

16. The cardboard in the figure is the net of a regular pyramid. PQRS is a square. It is given that VN = 30 cm and VQ = 34 cm.

- (a) Find the length of PQ.
 (b) Find the total surface area of the pyramid.

- (a) In $\triangle VQN$,
 $QN = \sqrt{VQ^2 - VN^2}$
 $= \sqrt{34^2 - 30^2} \text{ cm}$
 $= 16 \text{ cm}$
 $PQ = 2 \times QN = 2 \times 16 \text{ cm} = \underline{32 \text{ cm}}$

- (b) Total surface area of the pyramid
 $= 4 \times \text{area of } \triangle VPQ + \text{area of square PQRS}$
 $= \left(4 \times \frac{1}{2} \times 32 \times 30 + 32^2 \right) \text{ cm}^2$
 $= \underline{2944 \text{ cm}^2}$



PRE S3 MATHS L7 – AREA AND VOLUME (A)

17. The figure shows a food tent without base. It is in the shape of a right pyramid with square base $BCDE$.
If $BC = 60$ cm and $AB = 50$ cm, find the total surface area of the tent.

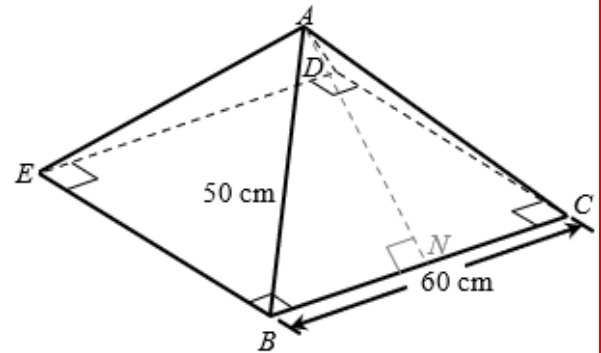
In $\triangle ABN$,

$$BN = \frac{1}{2} \times BC = \frac{1}{2} \times 60 \text{ cm} = 30 \text{ cm}$$

$$\begin{aligned} AN &= \sqrt{AB^2 - BN^2} \\ &= \sqrt{50^2 - 30^2} \text{ cm} \\ &= 40 \text{ cm} \end{aligned}$$

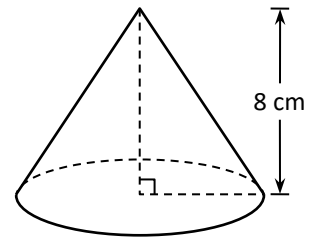
$$\begin{aligned} \text{Total surface area of the tent} &= 4 \times \text{area of } \triangle ABC \\ &= 4 \times \frac{1}{2} \times 60 \times 40 \text{ cm}^2 \\ &= \underline{4800 \text{ cm}^2} \end{aligned}$$

Let AN be the height of $\triangle ABC$.



18. In the figure, the volume of a right circular cone is $96\pi \text{ cm}^3$ and its height is 8 cm.

- Find the base radius of the circular cone.
- Find the slant height of the circular cone.
- Find the total surface area of the circular cone in terms of π .



(a) Let r cm be the base radius of the circular cone.

$$\frac{1}{3} \times \pi \times r^2 \times 8 = 96\pi$$

$$r^2 = 36$$

$$r = 6$$

$\therefore$ The base radius of the circular cone is 6 cm.

(b) Slant height of the circular cone $= \sqrt{6^2 + 8^2} \text{ cm}$
 $= \underline{10 \text{ cm}}$

(c) Total surface area of the circular cone
 $= (\pi \times 6 \times 10 + \pi \times 6^2) \text{ cm}^2$
 $= \underline{96\pi \text{ cm}^2}$

19. The base radius and height of a right circular cone are 11 cm and 60 cm respectively.

- Find the slant height of the circular cone.
- Find the total surface area of the circular cone in terms of π .

(a) Let ℓ cm be the slant height of the cone.

$$\begin{aligned} \ell &= \sqrt{11^2 + 60^2} \\ &= 61 \end{aligned}$$

$\therefore$ The slant height of the circular cone is 61 cm.

(b) Total surface area of the cone
 $= (\pi \times 11 \times 61 + \pi \times 11^2) \text{ cm}^2$
 $= \underline{792\pi \text{ cm}^2}$

20. **Fig. A** shows a glass filled up with apple juice. It is in the shape of an inverted right circular cone of base radius 9 cm and height 15 cm. All the juice in the glass is poured into an empty cylindrical vessel of base radius 6 cm (as shown in **Fig. B**). Find the depth of the juice in the cylindrical vessel.

Let d cm be the depth of apple juice in the cylindrical vessel.

Capacity of the glass = volume of the apple juice in the cylindrical vessel

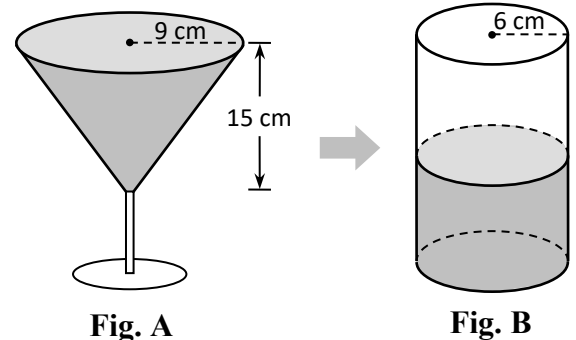
$$\frac{1}{3} \times \pi \times 9^2 \times 15 = \pi \times 6^2 \times d$$

$$405\pi = 36\pi d$$

$$d = \frac{405\pi}{36\pi}$$

$$= 11.25$$

$\therefore$ The depth of apple juice in the cylindrical vessel is 11.25 cm.



21. In the figure, the depth of the water in a cylindrical container is 8 cm. After a solid right circular cone with height 9 cm and the same base radius is put into the container, the water level rises to the top of the container. Find the height of the container.

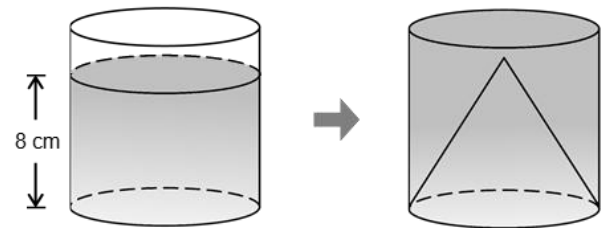
Let A cm² and h cm be the base area and the height of the container respectively.

$$A \times h = 8 \times A + \frac{1}{3} \times A \times 9$$

$$h = 8 + 3$$

$$= 11$$

$\therefore$ The height of the container is 11 cm.



22. Consider a regular pyramid A with a square base and a right circular cone B . It is given that the length of a side of the square base of pyramid A is the same as the base radius of cone B . If the two solids have the same volume, which solid will have a greater height? Explain your answer.

Let r cm be the length of a side of the square base of pyramid A and the base radius of cone B .

Let h_1 cm and h_2 cm be the heights of pyramid A and cone B respectively.

$\therefore$ Volume of pyramid A = volume of cone B

$$\frac{1}{3} \times r^2 \times h_1 = \frac{1}{3} \times \pi \times r^2 \times h_2$$

$$h_1 = \pi h_2$$

i.e. $h_1 > h_2$

$\therefore$ The height of pyramid A is greater.

23. A chocolate cuboid with dimensions $12.5 \text{ cm} \times 4 \text{ cm} \times 4 \text{ cm}$. It is melted and recast into some identical right circular cones of base radius 1.5 cm and height 2.4 cm. What is the maximum number of cones that can be made?

Let n be the number of cones that can be made.

$$12.5 \times 4 \times 4 = n \times \frac{1}{3} \times \pi \times 1.5^2 \times 2.4$$

$$200 = n \times 1.8\pi$$

$$n = 35.367\dots$$

$\therefore$ The maximum number of cones that can be made is 35.

24. **Fig. A** shows a paper cup in the shape of an inverted right circular cone.

- (a) Find the capacity of the paper cup.
(b) The paper cup is cut along PQ to form the sector in **Fig. B**. Find
(i) the area of the sector,
(ii) the angle of the sector.

(Give the answers correct to 3 significant figures.)

(a) Let r cm be the base radius of the paper cup.

$$r = \sqrt{20.5^2 - 20^2}$$

$$= 4.5$$

$\therefore$ Capacity of the paper cup

$$= \frac{1}{3} \times \pi \times 4.5^2 \times 20 \text{ cm}^3$$

$$= \underline{424 \text{ cm}^3}, \text{ cor. to 3 sig. fig.}$$

(b) (i) Area of the sector

= curved surface area of the paper cup

$$= \pi \times 4.5 \times 20.5 \text{ cm}^2$$

$$= 92.25\pi \text{ cm}^2$$

$$= \underline{290 \text{ cm}^2}, \text{ cor. to 3 sig. fig.}$$

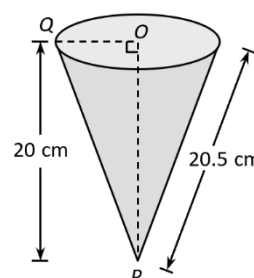


Fig. A

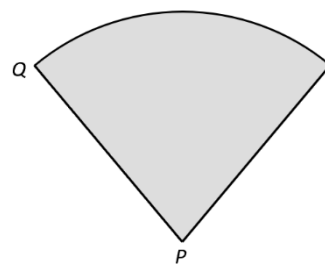


Fig. B

(ii) Let θ be the angle of the sector.

Radius of the sector

= slant height of the paper cup

$$= 20.5 \text{ cm}$$

$$\frac{\theta}{360^\circ} \times \pi \times 20.5^2 = 92.25\pi$$

$$\theta = 79.0^\circ, \text{ cor. to 3 sig. fig.}$$

$\therefore$ The angle of the sector is 79.0° .

25. **Fig. A** shows a sector. It is folded into an inverted right circular cone as shown in **Fig. B**.

(a) Find the base radius of the cone.

(b) Find the capacity of the cone.

(Give the answer correct to 3 significant figures.)

(a) Radius of the sector = slant height of the cone

$$= 12 \text{ cm}$$

Let r cm be the base radius of the cone.

$\therefore$ Curved surface area of the cone

= area of the sector

$$\therefore \pi \times r \times 12 = \frac{240^\circ}{360^\circ} \times \pi \times 12^2$$

$$r = 8$$

$\therefore$ The base radius of the cone is 8 cm.

Alternative

Let r cm be the base radius of the cone.

Circumference of the base of the cone = length of the arc in the sector

$$2\pi r = \frac{240^\circ}{360^\circ} \times 2\pi \times 12$$

$$r = 8$$

$\therefore$ The base radius of the cone is 8 cm.

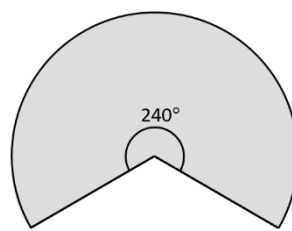


Fig. A

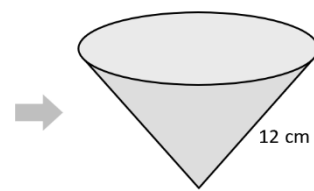


Fig. B

(b) Let h cm be the height of the cone.

$$h = \sqrt{12^2 - 8^2}$$

$$= \sqrt{80}$$

Capacity of the cone

$$= \frac{1}{3} \times \pi \times 8^2 \times \sqrt{80} \text{ cm}^3$$

$$= \underline{599 \text{ cm}^3}, \text{ cor. to 3 sig. fig.}$$